Zer0pa / ZPE-IoT

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Watch 0

ZPE-IoT V0.0: DETERMINISTIC IOT SENSOR CODEC: 1D Time-Series | Chemosense | 17x Mean Compression | Rust Core | PyO3 Native | Edge-Deployable

www.zer0pa.ai
MIT license
Code of conduct
Contributing
Security policy
0 stars 0 forks 0 watching Branches Activity Custom properties
Tags
Public repository

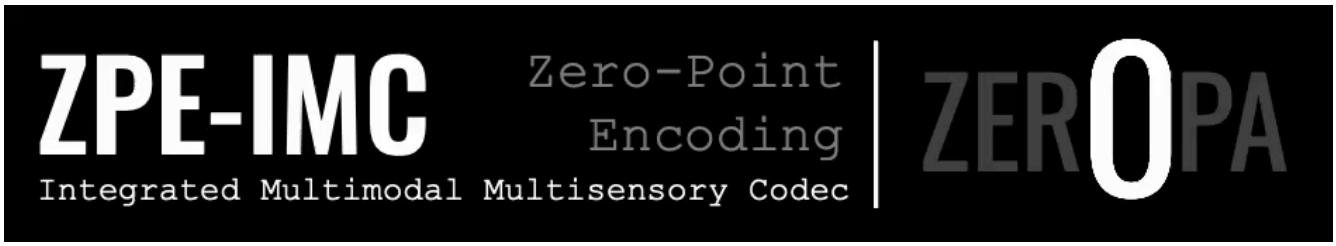
... 1 Branch 0 Tags Go to file Go to file Add file Code

Zer0pa-Architect-Prime feat: promote engineering closure onto main 7623f0d · 1 hour ago

.github	feat: promote engineering closure onto m...	1 hour ago
.gpd	feat: promote engineering closure onto m...	1 hour ago
c	[PHASE-3] SDK packaged, documentatio...	last month
core	feat: promote engineering closure onto m...	1 hour ago
docs	feat: promote engineering closure onto m...	1 hour ago
examples	[STEP-3.2] Remove generated example b...	last month
project_docs	feat: promote engineering closure onto m...	1 hour ago
proofs	feat: promote engineering closure onto m...	1 hour ago
python	feat: promote engineering closure onto m...	1 hour ago
release/RC_20260321T225526	docs: align authority surface and prune le...	1 hour ago
scripts	feat: promote engineering closure onto m...	1 hour ago
validation	feat: promote engineering closure onto m...	1 hour ago
.gitignore	feat: promote engineering closure onto m...	1 hour ago
AUDITOR_PLAYBOOK.md	docs: align authority surface and prune le...	1 hour ago
CHANGELOG.md	docs: align authority surface and prune le...	1 hour ago
CITATION.cff	docs: align release surface to March 21 au...	3 hours ago
CODE_OF_CONDUCT.md	Stage private repo and harden repo surface	2 weeks ago
CONTRIBUTING.md	docs: align authority surface and prune le...	1 hour ago
GOVERNANCE.md	docs: align authority surface and prune le...	1 hour ago
LANE_INTERLOCKS.md	Stage private repo and harden repo surface	2 weeks ago
LICENSE	[PHASE-1] Core codec complete, P0 gate ...	last month
PUBLIC_AUDIT_LIMITS.md	docs: align authority surface and prune le...	1 hour ago
README.md	docs: align authority surface and prune le...	1 hour ago
RELEASE_IMPACT.md	Stage private repo and harden repo surface	2 weeks ago

RELEASING.md	docs: align authority surface and prune le...	1 hour ago
ROADMAP.md	docs: align authority surface and prune le...	1 hour ago
SECURITY.md	docs: align authority surface and prune le...	1 hour ago
SUITE_COMPATIBILITY_VECTOR.json	Stage private repo and harden repo surface	2 weeks ago
SUPPORT.md	docs: align authority surface and prune le...	1 hour ago

README | Code of conduct | Contributing | MIT license | Security



ZPE-IoT

Docs | Proofs | API | Benchmarks | Release | Audit

Deterministic sensor compression and chemosense packetization for IoT time-series.

WHAT THIS IS

ZPE-IoT is a private-stage sensor compression SDK for 1D IoT time-series. The canonical implementation surface is the Rust core in `core/`, exposed through the nested Python distribution in `python/` and a repo-local PyO3 native build in `python/native/`.

This front door promotes only the March 21, 2026 repo-local authority surface. It does not claim public package availability, universal compressor dominance, or runtime coupling to ZPE-IMC.

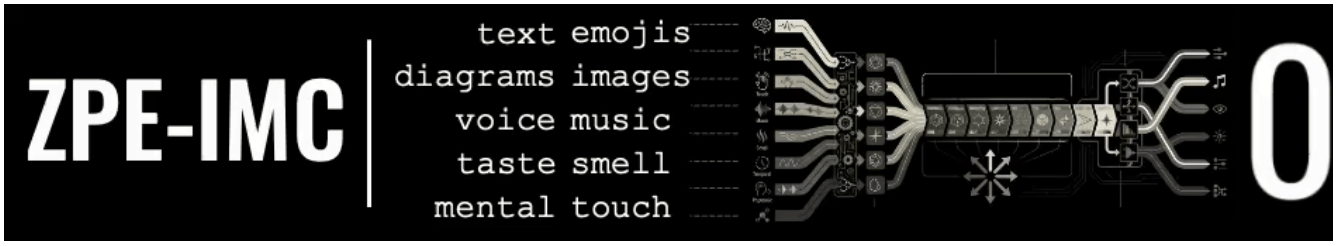
QUICKSTART AND AUTHORITY POINT

```
git clone https://github.com/Zer0pa/ZPE-IoT zpe-iot
cd zpe-iot
python -m pip install -e './python[dev]'
cargo test --manifest-path core/Cargo.toml --release
python validation/destruct_tests/run_all_dts.py --strict-gates
```

Field	Current truth	Evidence
As of	`2026-03-21`	Final status
Repository URL	`https://github.com/Zer0pa/ZPE-IoT`	Citation
Repo classification	`private-stage multi-surface codec repo`	Technical alignment proof
Release unit	`python/` distribution with bundled native wheel; `core/` and `c/` remain sibling engineering surfaces	Technical alignment proof
Acquisition surface	Private repo checkout or owner-shared built wheel from `python/dist/`	Native wheel verification
Managed preflight	`17 PASS / 0 FAIL / 1 DEFERRED`	Preflight report

Field	Current truth	Evidence
Strict DT	`27/27 PASS`	DT report
Fresh install smoke	`PASS` on local arm64 macOS cold install	Cold-install smoke
Benchmark authority	`E1` , `10/11 wins` , `17.16x` mean CR	E1 summary
Known real blockers	`none`	Technical alignment proof
Publication posture	`tag/index publication and outreach deferred pending explicit owner approval`	Preflight report
Canonical evidence entry	`proofs/PROOF_INDEX.md`	Proof index

`Managed preflight` is the build/install/release gate, `strict DT` is the destructive-test gate, and `E1` is the promoted real-public benchmark tier.



REPO SHAPE

Area	Purpose
`README.md` , `CHANGELOG.md` , `CONTRIBUTING.md` , `SECURITY.md` , `SUPPORT.md` , `GOVERNANCE.md` , `RELEASING.md` , `ROADMAP.md` , `CITATION.cff` , `LICENSE`	Root truth, governance, release, and citation surface
`python/`	Installable Python distribution, CLI, and package metadata
`python/native/`	Repo-local PyO3 native build surface used for bundled wheels
`core/`	Canonical Rust codec kernel and test surface
`docs/`	Reader-facing architecture, benchmark, support, and legal routing
`docs/family/`	IMC contract-alignment artifacts; documentary only, not runtime-coupled
`proofs/`	Current verdict, proof routing, receipts, runbooks, and artifacts
`validation/`	Datasets, benchmarks, destructive tests, and generated result JSON
`project_docs/` , `release/RC_*`	Operator lineage and historical release packets, not the front-door authority surface

OPEN RISKS (NON-BLOCKING)

Risk lens	Current state
Publication	Public package publication remains deferred by policy; use the private repo or owner-shared wheel instead of claiming PyPI/crates.io availability.
Benchmark boundary	The active E1 surface is `DS-01..DS-10` plus `DS-12` ; `DS-11` remains explicitly `BLOCKED` .
Comparator honesty	ZPE-IoT does not win every slice; `DS-12` is a competitor win on the current E1 real-public surface.
Native scope	Local arm64 macOS wheel install is verified; the multi-platform publish workflow exists but has not been executed as a public release event.
Fidelity boundary	ZPE-IoT is a bounded-lossy codec. It is not a fit for strict lossless reconstruction requirements.

CONTRIBUTING, SECURITY, SUPPORT

Route	Target
Documentation index	`docs/README.md`
Canonical doc registry	`docs/DOC_REGISTRY.md`
Architecture and runtime map	`docs/ARCHITECTURE.md`
API and CLI details	`docs/API.md` , `docs/CLI_CONTRACT.md`
Benchmark authority and boundaries	`docs/BENCHMARKS.md`
Audit replay path	`AUDITOR_PLAYBOOK.md`
Public audit boundary	`PUBLIC_AUDIT_LIMITS.md`
Contribution rules	`CONTRIBUTING.md`
Support routing	`docs/FAQ.md` first, then `docs/SUPPORT.md` , then `SUPPORT.md` for repo-level policy
Security reporting	`SECURITY.md`
Legal/release boundary	`docs/LEGAL_BOUNDARIES.md` , `RELEASING.md`

Releases

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 1

Zer0pa-Architect-Prime

Languages

Python 93.3% Rust 6.1% Other 0.6%